

Producing agri-food derived composts from coffee husk as primary feedstock at different temperature conditions

Author links open overlay

panelMarziyeh Hoseini ^a, [Stefania Cocco](#) ^a, Cristiano Casucci ^a, Valeria Cardelli ^a, [Maria Letizia Ruello](#) ^b, Dominique Serrani ^a, [Giuseppe Corti](#) ^{a c}

Show more

Add to Mendeley

Share

Cite

<https://doi.org/10.1016/j.jenvman.2024.123485>[Get rights and content](#)

Highlights

-
-

Coffee husk detoxifies from toxic compounds during the composting process.

-
-

Compost quality differs because of the diverse environmental temperatures.

-
-

Composts obtained under both conditions have moderate maturity.

-
-

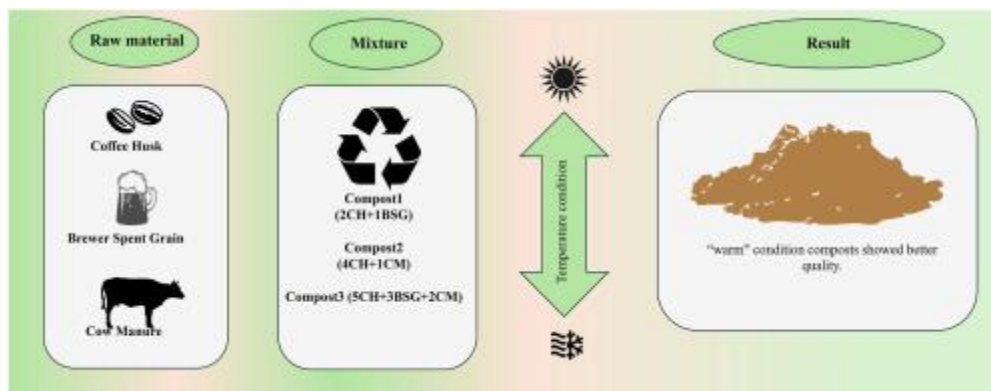
Compost obtained under “warm” condition showed better stability.

Abstract

There is a great global concern about agricultural wastes from food and feed [crop processing](#) that have significant environmental impacts. Composting is the most environmentally friendly, cost-effective, and efficient processes that can solve the problems of accumulation and toxicity of agricultural waste. The aim of this study is the detoxification of coffee husk by composting at two temperature conditions (“warm” and “cold”). In the greenhouse, the ambient temperature was changed day by day to mimic the situation of a spring to summer “warm” period ($\approx 16\text{--}34\text{ }^{\circ}\text{C}$) and a spring “cold” period ($\approx 7\text{--}20\text{ }^{\circ}\text{C}$) typical of central Italy. The coffee industry should accept the responsibility for the large amount of organic waste production, which presents toxicity and mass accumulation problems. Coffee husk as the main raw material is not used directly as bio-fertilizer

in [agriculture](#) sector due to the leaching of [phenolic compounds](#) and high pH value. The [brewing industry](#) is famous for its mass production, and the brewer residues as a by-product have an extremely acidic pH that makes them an unsuitable material for direct composting, but the mixture of these materials can optimize pH. The addition of cow manure accelerates [microbial activity](#) and is a strategy to improve composting rate and maturity. The following mixtures were tested: coffee husk and brewer spent grains in a proportion of 2:1 (Compost 1), coffee husk and cow manure in a proportion of 4:1 (Compost 2), and coffee husk, brewer spent grain, and cow manure in a proportion of 5:3:2 (Compost 3). Quality and maturity of the final composts appeared to be affected by the ambient temperature conditions, which remarkably affected pH, C/N ratio, nutrient and trace elements availability, germination index, [microbial biomass](#) carbon, and [FDA](#) hydrolysis. Results showed that both sets of temperatures produced composts to be considered standard compost, but "warm" conditions compost showed greater maturity, while the composts produced under "cold" conditions were able to increase seed germination.

Graphical abstract



1. [Download: Download high-res image \(300KB\)](#)

2. [Download: Download full-size image](#)

Introduction

Organic waste is considered a problem that is often solved by disposal or combustion. However, effective management and valorization of agricultural and food waste have positive results on resource conservation, environmental preservation, and economic growth (Preethi et al., 2024; Imran et al., 2022). Coffee is the most consumed beverage globally due to its strong aroma and stimulating effect. With over 85 known coffee species, *Coffea arabica* L. (known as Arabica coffee) and *Coffea canephora* L. (known as Robusta coffee) are the most known and consumed. Green coffee beans are rich in

phenolic compounds and antioxidants, primarily chlorogenic acids. During roasting, some phenolic and toxic compounds form, while others are decomposed by heat (Erskine et al., 2022). Coffee husk (CH) is the main by-product of coffee roasting process and, especially in developing countries, it is often disposed in large volumes without any pre-treatment (Hoseini et al., 2021). Because of its content of phenolic compounds and caffeine mutagenic contamination, disposal or distribution on soil surface poses serious risks as these molecules finally leak into the underground waters (Hoseini et al., 2021). Since the release of phenols and other toxic compounds in the environment has significant environmental impacts, including toxicity and potential carcinogenic effects on living organisms, pre-treatment of CH is crucial whatever its fate (Gami et al., 2015). The study of recycling CH by mixing it with materials efficient in removing most of the toxic compounds through composting is the main challenge of this study. There are different methods for removing toxic compounds. Conventional treatments such as distillation, absorption, extraction, chemical oxidation, and electrochemical oxidation show high efficiencies especially with various phenolic compounds. Advanced treatments such as Fenton processes, ozonation, wet air oxidation, and photochemical treatment use less chemicals than conventional ones, but they have high energy costs. Enzymatic treatment has proven to be an acceptable method, but is expensive. Instead, biological treatment such as composting is environmentally friendly and allows energy saving (Villegas et al., 2016).

Composting is the most convenient, environmentally friendly, and efficient method to remove toxic compounds of CH and manage its accumulation to improve waste management, ecosystem sustainability, and bio-fertilizers production. The composting process disinfects organic wastes from pathogens, eliminates most toxic molecules and weed seeds, and stabilizes C, N, and other nutrients in the organic fraction. Field application of compost can help maintain or increase soil organic matter content, biological activity, and porosity; therefore, water, air, and plant roots can penetrate the soil more easily, helping to reduce soil erosion and evapotranspiration, and contributing to land reclamation (Hoseini et al., 2021; Li et al., 2013).

CH characteristics, including a C/N ratio of 30 and its rich lignocellulose content, make it an ideal substrate for composting (Hoseini et al., 2021). Previous studies have compared the quality of composts made with CH alone with mixtures like cow manure, microbes, and fruit/vegetable wastes. They found that the mixtures provided the highest quality composts, improving soil fertility and crop yield while reducing toxic compounds (Dzung et al., 2013; Sekhar et al., 2014; Kassa and Workayehu, 2014). In earlier experience, we composted CH alone for three months, but the high pH (≈ 9) of the resulting compost suggested us to add other organic materials to optimize the final pH of the compost and help CH to further degrade and detoxify, as also reported by Ho et al. (2022). To investigate

the effect of ambient temperature on the quality of the final compost, we conducted an experiment using CH added of organic materials such as brewer spent grain (BSG) and cow manure (CM).

The brewing enterprise is characterized by a massive production of by-product. Following the basic principles of the circular economic system, several attempts to recycle BSG have been investigated. The literature review of BSG shows that it has about 75% moisture content, a C/N ratio around 17, and pH 5. These parameters are not suitable for producing high-quality compost. For this reason, Assandri et al. (2021) composted BSG by adding a lignocellulosic bulking agent (wheat straw, wood chips, sawdust) and farm animal manure to achieve a good composting process. Therefore, since the high amount of toxic compounds in CH makes the natural composting process time consuming, it has been suggested to add effective microorganisms and/or CM to improve the composting rate, stabilize organic materials, and increase the compost maturity (Rastogi et al., 2019).

There are studies that consider the effect of temperature on the amount of compost (Boldrin and Christensen, 2010) or on aeration of the compost pile (Vilela et al., 2022), but also studies that show the influence of factors such as climate, geography, and vegetation on the quality of the compost (Cao et al., 2023). However, there is no report about the effect of ambient temperature on the temperature of the composting piles and therefore, on the quality of the compost.

The aim of this work was therefore to assess CH detoxification via composting, with pH optimization through mixing CH with BSG and CM. At the same time, the effect of the ambient temperatures (“warm” and “cold”) on composts phytotoxicity and maturity was established by simulating the temperatures of a temperate area like that of central Italy.

Preparation of the piles and experimental design

CH, BSG, and CM as the raw materials were obtained from industrial plants of the Marche region, Italy. Since CH detoxification was the main objective of this study, CH was present in all mixtures. To optimize degradation of toxic CH molecules and the pH in BSG during composting, one mixture was with BSG. A second mixture was made with CM as an accelerator of the composting process and to increase the compost maturity. The third mixture was made of the three materials. The ratio of raw materials

Temperature monitoring during compost process

The temperatures of the composting piles recorded during the experimentation are reported in Fig. 1. In both ambient conditions, the temperature displayed an increment in the initial stage of composting from the 1st to 14th day, followed by a reduction in the later stages of composting. High temperature in the initial stages of composting helps to reduce

pathogenic microbes and to make mature compost (Ravindran et al., 2022). Combining the data of Table 1 and Fig. 1, it appeared that under

Conclusions

This study aimed at assessing compost maturity and related parameters. The quality and maturity of final composts were affected by ambient temperature conditions, which significantly affected pH, C/N ratio, nutrient content, trace elements availability, GI, MBC, and FDA hydrolysis. The obtained composts showed physicochemical and biological properties that allowed them to be considered as standard composts. However, composts obtained under “warm” conditions showed better quality, including more

CRediT authorship contribution statement

Marziyeh Hoseini: Writing – review & editing, Writing – original draft, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Stefania Cocco:** Writing – review & editing. **Cristiano Casucci:** Formal analysis. **Valeria Cardelli:** Formal analysis. **Maria Letizia Ruello:** Formal analysis. **Dominique Serrani:** Formal analysis. **Giuseppe Corti:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This research was funded through scholarship provided to the first author by the Università Politecnica delle Marche.